# Somerville Sample: $E = mc^2$

Here is some text *some italics* and **some bold** and ***some bold italics***
Comparisons
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$x < \mathbb{R} + \mathbb{T}$
A𝔸BB̋RR̋*R*
For $a = b = l$ $f' u' = g' = h' = u' = y'$ $a, b, c \in \mathbb{R}$, the identity $a^2 + b^2 = c^2$ sits alongside fractions $\frac{1}{2}$ and radicals $\sqrt{2}$. Nested scripts: $|x| = x_i^2 + x_{i_j}^{n+1} + e^{x^2}$.

## Fractions and radicals: $\sqrt{1 + x^2}$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \qquad \frac{1}{1 + \frac{1}{1+x}} \qquad \sqrt{1 + \sqrt{1 + \sqrt{x}}}$$

## Integrals, sums, and products: $\int_a^b f(x)\,\mathrm{d}x$

$$\int_0^1 x^2\,\mathrm{d}x = \frac{1}{3} \qquad \int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{p}$$

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2} \qquad \prod_{k=1}^{n} \frac{k+1}{k} = n + 1$$

## Delimiters and matrices: $\det \mathbf{A}$

$$\left(\frac{x+1}{x-1}\right)^2 \qquad \frac{u}{|u|}, \frac{v}{|v|} \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

## Limits, accents, and arrows: $\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = f'(x) \qquad \hat{f}(x) = \int_{-\infty}^{\infty} f(x)\, e^{-2pixx}\,\mathrm{d}x$$

$u' + v' = w'$ $\quad \overline{z} = x - iy$ $\quad x \mathbin{?} A \mathbin{?} B \iff x \mathbin{?} A \mathbin{?} x \mathbin{?} B$

# Integral sizes and italic correction: $\int_a^b$

Each row requests a larger integral at the same body size. Left: side limits. Right: limits above and below. The identical limit letters make horizontal shifts easier to compare.

$\int_a^a f(x)\,dx$

| | | |
|---|---|---|
| size 6pt | $\int_a^a f(x)\,\mathrm{d}x$ | $\int\limits_a^a f(x)\,\mathrm{d}x$ |
| size 9pt | $\int_a^a f(x)\,\mathrm{d}x$ | $\int\limits_a^a f(x)\,\mathrm{d}x$ |
| size 12pt | $\int_a^a f(x)\,\mathrm{d}x$ | $\int\limits_a^a f(x)\,\mathrm{d}x$ |
| size 18pt | $\int_a^a f(x)\,\mathrm{d}x$ | $\int\limits_a^a f(x)\,\mathrm{d}x$ |
| size 36pt | $\int_a^a f(x)\,\mathrm{d}x$ | $\int\limits_a^a f(x)\,\mathrm{d}x$ |
| size 48pt | $\int_a^a f(x)\,\mathrm{d}x$ | $\int\limits_a^a f(x)\,\mathrm{d}x$ |

# Miscellanea *mathematica*: $e^{ip} + 1 = 0$

Lorem ipsum dolor sit amet, $x \mathbin{?} \mathbb{R}$ consectetur adipiscing elit. Sed vestibulum $[0,1]$, ut $e > 0$ fermentum *aculis ornare*. Curabitur consequat $a_n = 1/(n+1)$ neque vitae sollicitudin; pellentesque habitant $b_n = (-1)^n a_n$ morbi tristique senectus et netus. Praesent $N \geq 1$, commodo viverra $s_N = \sum_{n=1}^{N} a_n$ quisque feugiat lobortis sapien. Nulla facilisi: etiam $d > 0$ tincidunt augue, vel $M < \infty$ porttitor massa, donec malesuada hendrerit arcu. Suspendisse finibus ipsum nec turpis sodales, non posuere justo dictum.

$$\int_0^1 x^n \,\mathrm{d}x = \frac{1}{n+1}, \qquad \sum_{n=0}^{\infty} \frac{z^n}{n!} = e^z, \qquad \lim_{n \to \infty} \left(1 + \frac{t}{n}\right)^n = e^t.$$

Vestibulum ante ipsum primis in faucibus, $f$ mollis nunc et $f'(a) \neq 0$ luctus sapien. Donec posuere $y = f(a) + f'(a)(x-a)$ ultricies ligula, vel *fermentum arcu dignissim sed*. Maecenas $R_2(h) = O(h^3)$ pretium lectus apud $h = 0$. Inter $a < b$ et $g \leq d$, sollicitudin ornare nibh, quisque consequat $\frac{p}{q}$ vitae facilisis $\sqrt{1 + \sqrt{1 + x^2}}$. Sed posuere, nunc et semper, consequat urna consectetur, aliquam erat volutpat, etiam porta tellus.

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{1}{2} \int_0^h (h-t)^2 f'''(a+t) \,\mathrm{d}t\,.$$

Integer pretium $(m, n) \mathbin{?} \mathbb{Z}^2$ vestibulum magna, dictum viverra $w_{m,n} = e^{-(m^2+n^2)}$ semper aliquam. Praesent commodo nunc vel nulla malesuada, et consectetur purus elementum. Hic $|z| < 1$, illic $\operatorname{Re}(s) > 1$, atque interdum sollicitudin $\prod_{k=1}^{N} (1 + u_k)$ mauris vitae risus accumsan. Phasellus $A \mathbin{?} B \mathbin{?} X$ porttitor ornare; $A \mathbin{?} B$, $A \mathbin{?} B$, et $X \mathbin{?} A$ habitant platea.

$$\sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} e^{-(m^2+n^2)} = \left( \sum_{k=-\infty}^{\infty} e^{-k^2} \right)^2, \qquad \sum_{k=0}^{N} z^k = \frac{1 - z^{N+1}}{1 - z}.$$

Aliquam erat volutpat. Sit $V = \mathbb{R}^3$, etiam $u, v \mathbin{?} V$, atque $?u, v? = \sum_{j=1}^{3} u_j v_j$ nunc vehicula. Nulla $|u| = \sqrt{?u, u?}$ facilisis eleifend, sed $t$ tempus fermentum. Vestibulum $A = (a_{ij})_{i,j=1}^{3}$ consequat nibh, ac malesuada lorem. Cum $A^{\mathrm{T}} = A$, dignissim ipsum; cum $A^{\mathrm{T}} A = I$, porttitor arcu. Mauris consequat sapien vitae turpis cursus, vestibulum blandit massa dapibus.

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2x_1 - x_2 \\ -x_1 + 2x_2 - x_3 \\ -x_2 + 2x_3 \end{pmatrix}.$$

Suspendisse potenti, $r(x) = e^{-x^2/2}/\sqrt{2p}$ faucibus mollis ante vestibulum. Etiam $m = 0$, semper $s^2 = 1$, et $r > 0$ tincidunt quam vitae posuere sapien. Nam $X$, $\mathbb{E}[X]$ et $\mathbb{P}(X \leq r)$

gravida ligula praesent. Donec dignissim blandit massa, ab $-\infty$ ad $+\infty$, commodo arcu pellentesque habitant morbi tristique senectus et netus.

$$\int_{-\infty}^{\infty} r(x)\,\mathrm{d}x = 1, \qquad \int_{-\infty}^{\infty} xr(x)\,\mathrm{d}x = 0, \qquad \int_{-\infty}^{\infty} x^2 r(x)\,\mathrm{d}x = 1.$$

Phasellus consectetur $\sin(nx)$ et $\cos(nx)$ convallis ornare. Vestibulum $\hat{f}(n)$ malesuada mauris, dum $e^{inx} = \cos(nx) + i\sin(nx)$ vulputate felis. Quisque $f(x) = \cos x + \frac{1}{2}\sin(2x)$ et $2p$; etiam $g(x) = e^{-x^2}$ urna sodales, finibus nisl vitae, porta neque. Ut enim ad minim veniam, cursus commodo facilisis consequat.

$$\hat{f}(n) = \frac{1}{2p}\int_{-p}^{p} f(x)\,e^{-inx}\,\mathrm{d}x, \qquad \frac{1}{2p}\int_{-p}^{p} |f(x)|^2\,\mathrm{d}x = \sum_{n\,?\,\mathbb{Z}} |\hat{f}(n)|^2 .$$

Nunc varius, $u = u(x,\,t)$ sollicitudin $?_t u = k\,?_x^2 u$ cum $k > 0$. Praesent tincidunt augue vel massa blandit, malesuada arcu semper $t \rightarrowtail \infty$. Etiam $?f$, interdum $Df$, atque $?^2 f/(?x\;?y)$ consequat nibh vestibulum. Aenean $u_0\,?\,L^1(\mathbb{R})$ finibus lorem, pellentesque sapien, dignissim semper magna.

$$u(x,\,t) = \frac{1}{\sqrt{4pkt}}\int_{-\infty}^{\infty} \exp\!\left(-\frac{(x-y)^2}{4kt}\right) u_0(y)\,\mathrm{d}y, \qquad t > 0.$$

Denique lorem ipsum, ornare minima: $x'$, $x''$, $x'''$, $a_{n+1}$, et $b_{i_j}^{k^2}$. Mauris $\overline{z}$, vestibulum $\hat{g}$, atque porttitor $\mathcal{X} \rightarrowtail \mathcal{Y}$ donec $\mathcal{Y} \rightarrowtail \mathcal{Z}$. Sive $p \rightrightarrows q$ vel $p \leftrightarrow q$, posuere ligula $0 < e\,?\,1$, malesuada cursus tellus, etiam finibus $\sqrt{a^2 + b^2} \leq |a| + |b|$.

## De campis electromagneticis: $? \times \mathbf{E} = -?_t \mathbf{B}$

Lorem ipsum Maxwellianum, sit $\mathbf{E} = \mathbf{E}(\mathbf{x},\,t)$ et $\mathbf{B} = \mathbf{B}(\mathbf{x},\,t)$ in spatio continuo. Vestibulum $r$, praesent $\mathbf{J}$, atque $c^{-2} = m_0 e_0$ consectetur adipiscing elit. Curabitur fluxus, nunc et semper, per superficies orientatas procedit; nulla facilisi, nam $? \cdot \mathbf{B} = 0$ dignissim magna et $?_t r + ? \cdot \mathbf{J} = 0$ consequat sapien. Donec malesuada, felis vel porttitor gravida, arcu erat sollicitudin lorem, vitae elementum risus neque sed justo.

$$? \cdot \mathbf{E} = \frac{r}{e_0} \qquad ? \cdot \mathbf{B} = 0$$
$$? \times \mathbf{E} = -\frac{?\mathbf{B}}{?t} \qquad ? \times \mathbf{B} = m_0\mathbf{J} + m_0 e_0 \frac{?\mathbf{E}}{?t}$$

Praesent integralia $?V$ et $?S$ circumveniunt, dum $\mathbf{n}$ normalis, d? tangentialis, et $\mathcal{Q}_V = \int_V r\,\mathrm{d}V$ massa verborum ornant. Aenean quis lacus vitae sapien bibendum cursus. Integer ut augue convallis, ornare arcu sed, dictum mauris; etiam $F_B$ interdum varius, sed $e_0 > 0$ semper constans.

$$\int_{?V} \mathbf{E}\cdot\mathbf{n}\,\mathrm{d}S = \frac{1}{e_0}\int_V r\,\mathrm{d}V, \qquad \oint_{?S} \mathbf{E}\cdot\mathrm{d}? = -\frac{\mathrm{d}}{\mathrm{d}t}\int_S \mathbf{B}\cdot\mathbf{n}\,\mathrm{d}S .$$

Pellentesque Poyntingiana, $\mathbf{S} = m_0^{-1}\mathbf{E} \times \mathbf{B}$ vehicula lucis, et $u$ condimentum energiae. Morbi tristique senectus et netus, $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ fermentum ipsum. Sed ut perspiciatis unde omnis, vestibulum $\mathbf{A}$ atque $f$ in faucibus orci luctus posuere.

$$u = \frac{e_0}{2}|\mathbf{E}|^2 + \frac{1}{2m_0}|\mathbf{B}|^2, \qquad \frac{?u}{?t} + ? \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}.$$

$$\mathbf{B} = ? \times \mathbf{A}, \qquad \mathbf{E} = -?f - \frac{?\mathbf{A}}{?t}, \qquad ?^2\mathbf{E} - \frac{1}{c^2}\frac{?^2\mathbf{E}}{?t^2} = \mathbf{0} \quad (r = 0,\ \mathbf{J} = \mathbf{0}).$$

## De undis et quantis: $\hat{H}q = Eq$

Suspendisse Schrödingeriana, $q$ ? $L^2(\mathbb{R}^3)$ volutpat magna, ? > 0 tincidunt minima. Nam $\hat{H}q = Eq$ commodo lectus, sed $\hat{\mathbf{p}} = -i? ?$ dignissim arcu. Vivamus ultricies, ipsum vitae malesuada tristique, urna erat consequat nibh, sit amet vulputate erat nulla non sem. Inter $?q, q? = 1$ et $|q|^2$ vestibulum probabilitatis, quisque elementum pretium sapien.

$$i?\frac{?q}{?t} = \left[-\frac{?^2}{2m}?^2 + V(\mathbf{x}, t)\right]q, \qquad \int_{\mathbb{R}^3}|q(\mathbf{x}, t)|^2\,\mathrm{d}^3\mathbf{x} = 1.$$

$$\left[\hat{x}, \hat{p}\right] = i?I, \qquad Dx\,Dp \geq \frac{?}{2}, \qquad E_n = ?w\left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, \cdots.$$

Aliquam Pauli matrices, $s = (s_1, s_2, s_3)$ ornare neque. Donec $s_j^2 = I$ tempus erat, $s_1 s_2 = is_3$ blandit justo, atque $\mathbf{a} \cdot s$ porttitor risus. Mauris at tellus et lectus facilisis, nec consequat nulla; curabitur convallis nisl id metus tempor, vitae fringilla neque sagittis. Nunc $\mathbf{a}, \mathbf{b}$ ? $\mathbb{R}^3$ semper interdum, et $\mathbf{a} \times \mathbf{b}$ vestibulum ipsum.

$$s_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad s_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

$$(\mathbf{a} \cdot s)\,(\mathbf{b} \cdot s) = (\mathbf{a} \cdot \mathbf{b})\,I + i(\mathbf{a} \times \mathbf{b}) \cdot s.$$

## De geometria et gravitate: $R_{mn} - \frac{1}{2}Rg_{mn}$

Aenean Einsteiniana, $g_{mn}$ vestibulum curvatum, et $R_{mn}$ sollicitudin magna. Proin $L$ ornare constans, $T_{mn}$ fermentum materiae, atque $?_m T^{mn} = 0$ consequat conservatum. Lorem ipsum dolor sit amet, consectetur adipiscing elit; morbi at arcu nec augue accumsan suscipit. Etiam $\mathrm{d}s^2 = g_{mn}\,\mathrm{d}x^m\mathrm{d}x^n$ interdum posuere, nec $t$ neque $l$ facilisis desunt.

$$R_{mn} - \frac{1}{2}Rg_{mn} + Lg_{mn} = \frac{8pG}{c^4}T_{mn}, \qquad E^2 = |\mathbf{p}|^2c^2 + m^2c^4.$$

$$\frac{\mathrm{d}^2 x^m}{\mathrm{d}t^2} + G^m_{ab}\frac{\mathrm{d}x^a}{\mathrm{d}t}\frac{\mathrm{d}x^b}{\mathrm{d}t} = 0, \qquad G^m_{ab} = \frac{1}{2}g^{mn}(?_a g_{nb} + ?_b g_{na} - ?_n g_{ab}).$$

## De fluidis et variationibus: $? \cdot \mathbf{u} = 0$

Vestibulum Navierianum, $\mathbf{u} = \mathbf{u}(\mathbf{x}, t)$ fluvius continuus, $p$ pretium pressionis, $n > 0$ viscositas serena. Nullam tincidunt libero sed ipsum faucibus, sed $? \cdot \mathbf{u} = 0$ sagittis incompressibilis. Etiam $r > 0$ constans et $\mathbf{f}$ sollicitudin externa, pellentesque habitant morbi tristique senectus. Donec at lectus vitae justo posuere maximus, ut $t \geq 0$ ornare nunc et semper.

$$\frac{?\mathbf{u}}{?t} + (\mathbf{u} \cdot ?)\,\mathbf{u} = -\frac{1}{r}?p + nD\mathbf{u} + \mathbf{f}, \qquad ? \cdot \mathbf{u} = 0.$$

Curabitur Lagrangiana, $L = L(\mathbf{q}, \dot{\mathbf{q}}, t)$ vestibulum motus, $S[\mathbf{q}]$ condimentum actionis. Sed $d\mathbf{q}(t_0) = d\mathbf{q}(t_1) = \mathbf{0}$ ornare fines, et $dS = 0$ praesent dictum. Praesent non ligula nec ipsum consequat sodales; maecenas $\mathbf{p} = ?L/?\dot{\mathbf{q}}$ fermentum neque, dum $H = \mathbf{p} \cdot \dot{\mathbf{q}} - L$ finibus arcu.

$$S[\mathbf{q}] = \int_{t_0}^{t_1} L(\mathbf{q}, \dot{\mathbf{q}}, t)\ \mathrm{d}t\,, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\frac{?L}{?\dot{q}_j} - \frac{?L}{?q_j} = 0.$$

$$\dot{q}_j = \frac{?H}{?p_j}, \qquad \dot{p}_j = -\frac{?H}{?q_j}, \qquad \{f, g\} = \sum\left(\frac{?f}{?q_j}\frac{?g}{?p_j} - \frac{?f}{?p_j}\frac{?g}{?q_j}\right).$$

## De numeris et infinitis: $\boldsymbol{z(2) = \dfrac{p^2}{6}}$

Lorem Eulerianum, $e^{ip} + 1 = 0$ dictum celebre; Riemanniana $z(s)$ porta infinita, ubi $\mathrm{Re}(s) > 1$ vestibulum absolutum. Quisque $p \in \mathbb{P}$ primus, $n \geq 1$ integer, et $G(s)$ mollis integralis. Fusce sit amet diam at nisi consequat egestas, vitae congue lorem suscipit. Donec $G(n + 1) = n!$ volutpat sapien, at $z(2) = p^2/6$ dignissim lectus.

$$z(s) = \sum_{n=1}^{\infty}\frac{1}{n^s} = \prod_{p\ \text{prime}}\frac{1}{1 - p^{-s}}\,, \qquad G(s) = \int_0^\infty t^{s-1}e^{-t}\,\mathrm{d}t\,.$$

$$G(s)\,G(1 - s) = \frac{p}{\sin ps}, \qquad n! \sim \sqrt{2pn}\left(\frac{n}{e}\right)^n.$$

Morbi Cauchyana, $f$ holomorpha, $g$ curva positiva, et $z_0$ punctum interius. Vestibulum ante ipsum primis, $f^{(n)}(z_0)$ sollicitudin alta, $n!$ consequat frequens. Sed ut perspiciatis, finibus neque sed, blandit massa; $\oint_g f(z)\,\mathrm{d}z$ circumdat margines, nec $2pi$ nec $z - z_0$ memoria deficit.

$$f^{(n)}(z_0) = \frac{n!}{2pi}\oint_g \frac{f(z)}{(z-z_0)^{n+1}}\,\mathrm{d}z\,, \qquad \int_{-\infty}^{\infty} \frac{\mathrm{d}x}{1+x^2} = p.$$

## De probabilitate et informatione: $\mathbb{E}[\mathbf{X}] = m$

Suspendisse Bayesiana, $\mathbb{P}(B) > 0$ porta aperta, $\mathbb{P}(A \mathbin{?} B)$ condimentum novum. Praesent $p_i > 0$, $q_i > 0$, et $\sum_i p_i = \sum_i q_i = 1$ ornare simplex. Cras consectetur sapien vel urna commodo, interdum $t \mathbin{?} \mathbb{R}^d$ fermentum variabile, interdum $?_t?$ sagittis ascendens. Mauris accumsan lorem ut magna pretium, in placerat massa dictum.

$$\mathbb{P}(A \mathbin{?} B) = \frac{\mathbb{P}(B \mathbin{?} A)\,\mathbb{P}(A)}{\mathbb{P}(B)}, \qquad H(p) = -\sum_i p_i \log p_i\,, \qquad D(p|q) = \sum_i p_i \log \frac{p_i}{q_i} \geq 0.$$

Denique Gaussiana, $\mathbf{x}, m \mathbin{?} \mathbb{R}^d$, et $S$ symmetrica positiva. Pellentesque $\det S > 0$ dignissim plena, dum $(\mathbf{x} - m)^{\mathrm{T}}$ vestibulum transpositum. Lorem ipsum dolor sit amet, consectetur adipiscing elit; etiam dictum magna, porta serena, finibus purus.

$$p(\mathbf{x}) = \frac{\exp\!\left[-\frac{1}{2}(\mathbf{x}-m)^{\mathrm{T}} S^{-1} (\mathbf{x}-m)\right]}{(2p)^{d/2}\sqrt{\det S}}.$$

## De fractionibus continuis et radicibus: $\sqrt[n]{1+x^n}$

Lorem ipsum profundum, $\frac{1}{1+\frac{1}{1+x}}$ vestibulum angustum, $\sqrt{\frac{a_i^2+b_i^2}{1+c_i^2}}$ porttitor altum. Praesent $x^{\frac{1}{2}}, x_{\frac{1}{2}}$, atque $e^{-\frac{x^2}{2s^2}}$ sollicitudin minuta. Curabitur superius et inferius, cras posuere nibh; nam $\frac{\sqrt{1+x^2}}{\sqrt{1-x^2}}$ inter verba procedit.

$$v = 1 + \frac{1}{1+\frac{1}{1+\frac{1}{1+?}}}, \qquad \sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}, \qquad \sqrt[n+1]{\frac{1+x^{2n+2}}{1+x^2}}.$$

$$\frac{\displaystyle\sum_{j=1}^{n} \frac{a_j}{\sqrt{1+b_j^2}}}{1+\displaystyle\int_0^1 \frac{\sqrt{1+t^4}}{1+t^2}\,\mathrm{d}t} \qquad \sqrt{\frac{1+\frac{x^2}{1+y^2}}{1+\frac{y^2}{1+x^2}}} \qquad \sqrt[3]{\sqrt[5]{\sqrt[7]{1+z^{105}}}}.$$

# De indicibus et litteris inclinatis: $\mathbf{T}_{i_j}^{k^{?}}$

Vestibulum $f_{i}$, $f^{j}$, $f_{i}^{j}$, $f_{i_{k}}^{j^{2}}$ et $f_{i_{j_{k}}}^{n^{m^{2}}}$ ornare creberrimum. Aliquam $\mathcal{F}_{a}^{b}$, $\mathcal{I}_{a}^{b}$, $V_{a}^{b}$, $W_{a}^{b}$, $G_{a}^{b}$ et $\int_{a}^{b}$ finibus obliquis; $f^{2\prime}$, $f_{i}^{\prime\prime}$, $f^{\prime\prime\prime}(x)$, $x_{n}^{\prime}$ semper praesent. Donec $\mathbf{A}_{i_{j}}^{k^{?}}$ nec $S_{a_{b}}^{g^{d}}$ facilisis desunt, in hac habitasse platea dictumst.

$$
{}_{6}^{14}\mathrm{C} \qquad {}_{92}^{238}\mathrm{U} \qquad {}_{b}^{a}\mathbf{T}_{g}^{d} \qquad R_{smn}^{r} = ?_{m}G_{ns}^{r} - ?_{n}G_{ms}^{r} + G_{ml}^{r}G_{ns}^{l} - G_{nl}^{r}G_{ms}^{l}.
$$

$$
\sum_{i=1}^{n} a_{i}^{\sum_{j=1}^{m} b_{ij}} \qquad e^{-\frac{x^{2}}{2s^{2}}} \qquad \frac{\mathrm{d}^{n}}{\mathrm{d}x^{n}} e^{-x^{2}} \Big|_{x=0} \qquad \left( \frac{x_{i}^{j_{k}}}{y_{m_{n}}^{p^{q}}} \right)^{2}.
$$

# De limitibus multiplicibus: $\sum_{i,j=1}^{n} a_{ij}$

Integer $\sum_{i=1}^{n}$, $\prod_{j=1}^{m}$, atque $\int_{0}^{1}$ vehicula parva; sed in spatio amplo ${}_{a?A}\, U_{a}$ et ${}_{n\geq 1}\, F_{n}$ porttitor magna. Praesent $\lim_{n \rightarrowtail \infty}$ volutpat lenta, $\sup_{x?K} |f(x)|$ dignissim finita.
Etiam $\inf_{y?Y} d(x, y)$ sollicitudin propinqua.

$$
\sum_{\substack{1 \leq i,\, j \leq n \\ i \neq j}} a_{ij} \qquad \prod_{\substack{p \leq N \\ p?\mathbb{P}}} \left( 1 - \frac{1}{p^{s}} \right)^{-1} \qquad \lim_{\substack{n \rightarrowtail \infty \\ h \rightarrowtail 0^{+}}} \frac{f_{n}(x+h) - f_{n}(x)}{h}.
$$

$$
\overset{\infty}{\underset{n=1}{}} \overset{\infty}{\underset{m=n}{}} A_{m} \; ? \; \overset{\infty}{\underset{n=1}{}} \overset{\infty}{\underset{m=n}{}} A_{m}, \qquad \left| \sum_{j=1}^{n} \mathbf{v}_{j} \right|^{2} = \sum_{i,j=1}^{n} ?\mathbf{v}_{i}, \mathbf{v}_{j}? \; .
$$

# De integralibus altissimis: $\int_{M} \mathrm{d}u$

Aenean Stokesiana, $w$ forma lenis, $M$ varietas orientata, et $?M$ terminus ordinatus. Nulla facilisi: $\mathrm{d}^{2}w = 0$, $? \times ?f = \mathbf{0}$, et $? \cdot (? \times \mathbf{A}) = 0$ semper concordant. Sed $\int_{M} \mathrm{d}w = \int_{?M} w$ vestibulum antiquum; praesent limites superiores atque inferiores, commodo arcu, finibus serenis.

$$
\int_{0}^{1} \frac{1}{\sqrt{1 + \frac{x^{2}}{1+x^{4}}}} \, \mathrm{d}x \qquad \int_{0}^{1} \frac{1}{\sqrt{1 + \frac{x^{2}}{1+x^{4}}}} \, \mathrm{d}x \, .
$$

$$\int_{-\infty}^{+\infty} \left[ \frac{\int_0^1 e^{-x^2t^2}\,\mathrm{d}t}{1+\frac{x^2}{1+x^2}} \right] e^{-x^2}\,\mathrm{d}x \qquad \int_{-\infty}^{+\infty} \frac{e^{-x^2}}{\sqrt{1+x^2}}\,\mathrm{d}x\,.$$

$$\iint_S (? \times \mathbf{F}) \cdot \mathbf{n}\,\mathrm{d}S = \oint_{?S} \mathbf{F} \cdot \mathrm{d}?\,, \qquad \iiint_V ? \cdot \mathbf{F}\,\mathrm{d}V = \iint_{?V} \mathbf{F} \cdot \mathbf{n}\,\mathrm{d}S\,.$$

## De accentibus et vinculis: $\hat{A}\hat{B}$

Mauris $\hat{f}$, $\hat{f}\hat{g}$, $\hat{f}\hat{g}\hat{h}$, $x\hat{y}z$, et $\overline{z_1 + z_2}$ condimentum superpositum. Vestibulum $\hat{x}$, $\hat{x}$, $\hat{\mathbf{x}}$, $\hat{\mathbf{x}}$, atque $\hat{f}$ ornare duplex. Phasellus volutpat $\hat{f}_i$ tincidunt ardua, nec $\bar{\bar{z}}$ finibus obscuris.

$$\hat{x} \quad x\hat{y} \quad x\hat{y}z \quad x_1 + x_2 \,\hat{?}\, x_3 + x_4 \qquad \mathbf{A}\hat{\mathbf{B}}\mathbf{C} \quad \overline{\frac{z_1 + z_2}{1 + z_1 z_2}}.$$

$$x_1 + x_2 \overset{n}{\hat{?}} ? + x_n = x + \overset{?}{\underset{n}{\hat{?}}} + x \quad (x_j = x), \qquad a \overset{u}{\hat{?}} b + c \overset{v}{\hat{?}} d. \overset{?}{\underset{u+v}{\hat{?}}}$$

$$\overset{\rightarrow}{AB} + \overset{\rightarrow}{BC} = \overset{\rightarrow}{AC}, \qquad \overline{z_1 \overset{?}{\hat{?}} ? + z_n} = \overline{z_1} + ? + \overline{z_n}.$$

## De parenthesibus et matricibus magnis: $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$

Curabitur ?$\mathbf{x}$, $\mathbf{y}$? spatium internum, $\lfloor x \rfloor$ gradus inferior, $\lceil x \rceil$ superior. Etiam $\{x \,?\, \mathbb{R} : x^2 < 2\}$ habitant aperte, $|\det \mathbf{A}|$ dignissim absoluta. In hac habitasse platea dictumst, cras $\mathbf{A}^{-1}$ sollicitudin inversa, sed $\det \mathbf{A} \neq 0$ vestibulum necessarium.

$$\left( \frac{1}{1+x^2}, \frac{x}{1+x^2} \right), \left( \sqrt{\frac{1+y}{1-y}}, \sqrt{\frac{1-y}{1+y}} \right) \qquad \left| \frac{\sum_{k=1}^{n} k^2}{\sum_{k=1}^{n} k} \right|.$$

$$\mathbf{H}_4 = \begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{pmatrix}, \qquad \det \begin{pmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{pmatrix} = (y-x)\,(z-x)\,(z-y)\,.$$

$$\left\{ \left[ \left( \frac{\int_0^1 t^{a-1}(1-t)^{b-1}\,\mathrm{d}t}{\frac{G(a)\,G(b)}{G(a+b)}} \right) \right]^2 \right\}^{1/2} = 1, \qquad a, b > 0.$$

# De casibus et aequationibus ordinatis: $d_{ij}$

Lorem ipsum distributionum, $d_{ij}$ porttitor discretum, $d(x)$ sollicitudin singularis. Praesent $\int_{\mathbb{R}} d(x-a)\,f(x)\,\mathrm{d}x = f(a)$ volutpat concentrata, dum $H'(x) = d(x)$ sensu distributionum. Vivamus at $x = 0$ posuere media, etiam $x < 0$ et $x > 0$ facilisis opposita. Donec ultricies sapien vitae massa cursus, interdum condimentum tellus, semper vestibulum purus.

$$H(x) = \begin{cases} 0 & x < 0, \\ \frac{1}{2} & x = 0, \\ 1 & x > 0, \end{cases} \qquad d_{ij} = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

$$\begin{aligned} |\mathbf{x} + \mathbf{y}|^2 &= ?\mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y}? \\ &= |\mathbf{x}|^2 + 2?\mathbf{x}, \mathbf{y}? + |\mathbf{y}|^2 \\ &\leq |\mathbf{x}|^2 + 2|\mathbf{x}|\,|\mathbf{y}| + |\mathbf{y}|^2 \\ &= (|\mathbf{x}| + |\mathbf{y}|)^2. \end{aligned}$$

# De formis exterioribus et identitatibus: $\mathrm{d}^2 \varpi = 0$

Aenean $\varpi = \sum_i a_i\,\mathrm{d}x^i$ vestibulum covarians, $h = \sum_j b_j\,\mathrm{d}x^j$ dignissim alternans. Morbi $\varpi \,?\, h = -h \,?\, \varpi$ commodo antisymmetricum, et $\mathrm{d}^2 = 0$ dictum breve. Nullam F, L, H, et $\mathbb{C} \,?\, \mathbb{H}$ ornare varia; $\mathbf{a} \,?\, \mathbf{b}$ condimentum tensoriale.

$$\sum \bigcirc_{i=1}^{n} \frac{\bigotimes a_n}{\bigodot b_n} \neq \bigcirc \prod_{i=1}^{n} \frac{\sum a_n}{\bigoplus b_n}$$

Denique $\frac{\int_0^1 f(x)\,\mathrm{d}x}{\sqrt{1+a^2}}$ inter verba surgit, dum $\frac{a_i^2}{1+b_j^2}$ minuitur, et $x_{ij}^{k^?}$ paulatim recedit. Lorem ipsum dolor sit amet, consectetur adipiscing elit; $\mathbf{0}$, $\mathbf{I}$, $a$, $b$, et $?$ vestibulum novissimum. Quisque $\left|\frac{\mathbf{x}}{|\mathbf{x}|}\right| = 1$ finibus clausis, sollicitudin integra, commodo sereno.

# De partitionibus et spectris: $\mathcal{Z}_N(b)$

Vestibulum harmonicum, $b > 0$, $\omega_j > 0$, et $E_{\mathbf{n}} = \sum_{j=1}^{N} ?\omega_j (n_j + \tfrac{1}{2})$ fermentum discretum. Praesent $\mathbf{n} ? \mathbb{N}_0^N$ porttitor occupationis; quisque modus suam frequentiam retinet, dum $\mathcal{Z}_N(b)$ consequat multiplicativa colligit.

$$\mathcal{Z}_N(b) = \sum_{\mathbf{n} ? \mathbb{N}_0^N} \exp\left[ -b \sum_{j=1}^{N} ?\omega_j \left( n_j + \frac{1}{2} \right) \right] = \prod_{j=1}^{N} \frac{e^{-b ?\omega_j / 2}}{1 - e^{-b ?\omega_j}} \, .$$

Mauris $\mathbf{A} ? \mathbb{C}^{N \times N}$ vestibulum spectrale, $l_1, \dots, l_N$ radices ordinatae. Nam $|z| \, |\mathbf{A}| < 1$ porta convergentiae, et $\operatorname{tr}(\mathbf{A}^r) = \sum_j l_j^r$ dictum invarians. Integer finibus purus, curabitur series concordes, donec determinantia in exponentia conveniunt.

$$\det(\mathbf{I} - z\mathbf{A})^{-1} = \prod_{j=1}^{N} \frac{1}{1 - z l_j} = \exp\left[ \sum_{r=1}^{\infty} \frac{z^r}{r} \operatorname{tr}(\mathbf{A}^r) \right].$$